

Assignment 4

1. Let $A_{m,n}$ be from the proof of Egorov's Theorem. Prove that f_k converging pointwise to f is equivalent to $\bigcup_{m=1}^{\infty} A_{m,n} = X$ for each $n \in \mathbb{N}$.

Proof. ($\implies$) Note that we can replace $\frac{1}{n}$, $n \in \mathbb{N}$ with $\varepsilon > 0$ using the Archimedean property. Suppose f_k converges pointwise. $A_{m,n} = \bigcap_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| < \varepsilon\}$. We let $E_k = \{x \in X : |f_k(x) - f(x)| < \varepsilon\}$. Thus, each E_k is the set of all points of X such that the k^{th} element of $\{f_n\}_{n \in \mathbb{N}}$ is within ε of f at x . Thus, $\bigcap_{k=m}^{\infty} E_k$ is the set of all points of X such that **all** functions f_k , $k \geq m$ are within ε of f at x . By definition of pointwise convergence, for all $x \in X, \varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $|f_k(x) - f(x)| < \varepsilon$ for all $k \geq n$. We now show that all $x \in X$ must be in $\bigcup_{m=1}^{\infty} [\bigcap_{k=m}^{\infty} E_k]$. Let $x \in X$. By definition of pointwise convergence, there must exist $n \in \mathbb{N}$ to make $|f_k(x) - f(x)| < \varepsilon$ whenever $k \geq n$. Thus:

$$x \in \bigcap_{k \geq n} E_k \subseteq \bigcup_{m \in \mathbb{N}} \bigcap_{k=m}^{\infty} E_k = \bigcup_{m=1}^{\infty} A_{m,n}.$$

Because x was chosen arbitrarily, this is true for all $x \in X$. Thus, we have shown that $X \subseteq \bigcup_{m=1}^{\infty} A_{m,n}$. Because all elements of $\bigcup_{m=1}^{\infty} A_{m,n}$ must be from X , by definition, $\bigcup_{m=1}^{\infty} A_{m,n} \subseteq X$. Because $X \subseteq \bigcup_{m=1}^{\infty} A_{m,n}$, and $\bigcup_{m=1}^{\infty} A_{m,n} \subseteq X$, $\bigcup_{m=1}^{\infty} A_{m,n} = X$. ($\impliedby$) Now suppose $\bigcup_{m=1}^{\infty} A_{m,n} = X$ for each $n \in \mathbb{N}$. This is equivalent to the following by the Archimedean property:

$$\bigcup_{m=1}^{\infty} \bigcap_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| < \varepsilon\} = X, \text{ for all } \varepsilon > 0.$$

We now show that this implies that $\{f_k\}_{k \in \mathbb{N}}$ is pointwise convergent on X . Let $x \in X$. Thus, $x \in \bigcup_{m=1}^{\infty} \bigcap_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| < \varepsilon\}$. Because this is a union, x must be within one of the elements of the union, thus $x \in \bigcap_{k=l}^{\infty} \{x \in X : |f_k(x) - f(x)| < \varepsilon\}$, $l \in \mathbb{N}$. This implies that for all $k \geq l$, $|f_k(x) - f(x)| < \varepsilon$. Therefore, we have:

$$\text{Given } x \in X, \varepsilon > 0, \exists l \in \mathbb{N} \text{ s.t. } \forall k \geq l, |f_k(x) - f(x)| < \varepsilon.$$

Notice that this is the definition of pointwise convergence over X . □

2. (a) Let $(X, \mathcal{S}, \mu)$ be a measure space and assume that $\mu(X) < \infty$. Let $f_1, f_2, f_3, \dots$ be a sequence of measurable functions on X converging to some function f pointwise. Prove that for all $\alpha > 0$

$$\lim_{n \rightarrow \infty} \mu(\{x \in X : |f_n(x) - f(x)| \geq \alpha\}) = 0$$

Proof. From problem 1, the definition of pointwise convergence implies:

$$\bigcup_{m=1}^{\infty} \bigcap_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| < \alpha\} = X.$$

We now take the complement of both sides to get:

$$\begin{aligned} & \left[\bigcup_{m=1}^{\infty} \bigcap_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| < \alpha\} \right]^C = \emptyset. \\ \iff & \bigcap_{m=1}^{\infty} \left[\bigcap_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| < \alpha\} \right]^C = \emptyset. \\ \iff & \bigcap_{m=1}^{\infty} \bigcup_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| \geq \alpha\} = \emptyset. \\ \implies & \mu\left(\bigcap_{m=1}^{\infty} \bigcup_{k=m}^{\infty} \{x \in X : |f_k(x) - f(x)| \geq \alpha\}\right) = \mu(\emptyset) = 0. \end{aligned}$$

Also, notice that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \mu(\{x \in X : |f_n(x) - f(x)| < \alpha\}) &= \\ & \mu\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \{x \in X : |f_k(x) - f(x)| < \alpha\}\right). \end{aligned}$$

and equivalently,

$$\begin{aligned} \lim_{n \rightarrow \infty} \mu(\{x \in X : |f_n(x) - f(x)| \geq \alpha\}) &= \\ & \mu\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{x \in X : |f_k(x) - f(x)| \geq \alpha\}\right). \end{aligned}$$

by set complements. Therefore,

$$\lim_{n \rightarrow \infty} \mu(\{x \in X : |f_k(x) - f(x)| \geq \alpha\}) = \mu(\emptyset) = 0.$$

□

- (b) Give an example of a sequence of measurable functions $f_1, f_2, f_3, \dots$ defined on $\mathbb{R}$ with Lebesgue measure such that f_n converges pointwise to some measurable function f and that there exists $\alpha > 0$ with

$$\lim_{n \rightarrow \infty} |\{x \in \mathbb{R} : |f_n(x) - f(x)| \geq \alpha\}| \neq 0$$

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as follows:

$$f = \begin{cases} 1 & x \in \mathbb{N} \\ 0 & x \notin \mathbb{N} \end{cases}$$

And the sequence $\{f_n\}_{n \in \mathbb{N}}$ defined as follows:

$$f_n = \begin{cases} 1 & (x - \frac{1}{n}, x + \frac{1}{n}) \cap \mathbb{N} \neq \emptyset \\ 0 & (x - \frac{1}{n}, x + \frac{1}{n}) \cap \mathbb{N} = \emptyset \end{cases}$$

Clearly, this sequence converges pointwise to f and:

$$|\{x \in \mathbb{R} : |f_n(x) - f(x)| \geq 1\}| = \sum_{k=1}^{\infty} \frac{2}{n} = \infty \text{ for all } n.$$

Thus, $\lim_{n \rightarrow \infty} |\{x \in \mathbb{R} : |f_n(x) - f(x)| \geq 1\}| = \infty$.

3. Prove the following generalization of Egorov's theorem:

Let $(X, \mathcal{S}, \mu)$ be a measure space and assume that $\mu(X) < \infty$. Let $f_1, f_2, f_3, \dots$ be a sequence of measurable functions on X and $f : X \rightarrow \mathbb{R}$ a function. Let

$$B = \{x \in X : \lim_{k \rightarrow \infty} f_k(x) \neq f(x)\}$$

and assume that $\mu(B) = 0$. Then for every $\epsilon > 0$ there exists $E \in \mathcal{S}$ such that $\mu(X \setminus E) < \epsilon$ and $\{f_k\}_{k=1}^\infty$ converges to f uniformly on E .

Proof. Let $\Xi = X \setminus B$. $X \setminus B$ and B are disjoint, (because $B \subseteq X$) thus:

$$\mu(X) = \mu(X \setminus B) + \mu(B) = \mu(X \setminus B) + 0 = \mu(X \setminus B).$$

Therefore $\mu(X) = \mu(X \setminus B)$. We now apply Egorov's theorem on the set Ξ , to find a set $E \subseteq \Xi$, with $\{f_k\}_{k \in \mathbb{N}}$ uniformly convergent on E and $\mu(\Xi \setminus E) < \epsilon$ for all $\epsilon > 0$. Now all we must show is that $\mu(X \setminus E) = \mu(\Xi \setminus E) < \epsilon$. Because $E \subseteq X$:

$$\begin{aligned} \mu(X \setminus E) &= \mu(X) - \mu(E). \\ &= \mu(\Xi) - \mu(E). \\ &= \mu(\Xi \setminus E). \quad (\text{because } E \subseteq \Xi) \\ &< \epsilon. \end{aligned}$$

We have found E where $\{f_k\}_{k \in \mathbb{N}}$ is uniformly convergent on E , and $\mu(X \setminus E) < \epsilon$. $\square$

4. Let $(X, \mathcal{S}, \mu)$ be a measure space. Let $E \subseteq X$ and $\mu(E) = 0$. Show that if $f : E \rightarrow [0, \infty]$ is $\mathcal{S}$ -measurable then

$$\int_E f d\mu = 0.$$

Proof. let $\varphi : X \rightarrow \mathbb{R}$ be a simple function ($\varphi = \sum_{k=1}^n c_k \chi_{E_k}$), where $\varphi \leq f$. We know that:

$$\int_E \varphi d\mu = \int_E \left[\sum_{k=1}^n c_k \chi_{E_k} \right] d\mu = \sum_{k=1}^n c_k \mu(E_k).$$

But, because $\mu(E) = 0$, by ordering we know that any subset of E must have measure less or equal to $\mu(E)$. Thus, all E_k have measure 0. Therefore:

$$\sum_{k=1}^n c_k \mu(E_k) = \sum_{k=1}^n c_k (0) = 0.$$

We now apply the monotone convergence theorem, so:

$$\begin{aligned} \int_E f d\mu &= \sup \left\{ \sum_{k=1}^n c_k \mu(E_k) \right\} \\ &= \sup \{0\} \\ &= 0. \end{aligned}$$

□